Task 1.10

Project Reflections:

From Data to Action: Optimising Marketing Through Regional Sales Analysis

Overview:

To verify whether GameCo's assumption that regional sales patterns have remained stable over time, I conducted a detailed analysis of the dataset containing game releases, sales, and regional performance. My goal was to identify trends in market dominance, volatility, and growth across North America, Europe, and Japan, providing actionable insights for marketing budget allocation.

Data Grouping and Summarisation:
I began by grouping the dataset using a PivotTable. The **Year** field was placed in the Rows area and the **Name** field in the Values area, set to Count, to track the number of games released per year. **Platform**, **Genre**, and **Publisher** were included in the Filters area, allowing flexible examination of subsets without altering the raw data. Years with very low row counts (e.g., 1977–1980, 2017, 2020) were excluded to avoid skewing trends due to incomplete reporting.
To analyse sales patterns, I summarised **global sales by region over time** and **calculated each region's share of total sales**. This allowed me to compare **North America**, **Europe**, and **Japan** across decades. Visualising the data with line charts made the trends immediately apparent: *North America led early markets*, *Japan surged in the late 1980s–1990s*, and *Europe gradually caught up*. In recent years, **all three regions contribute roughly equally (~30% each)**.

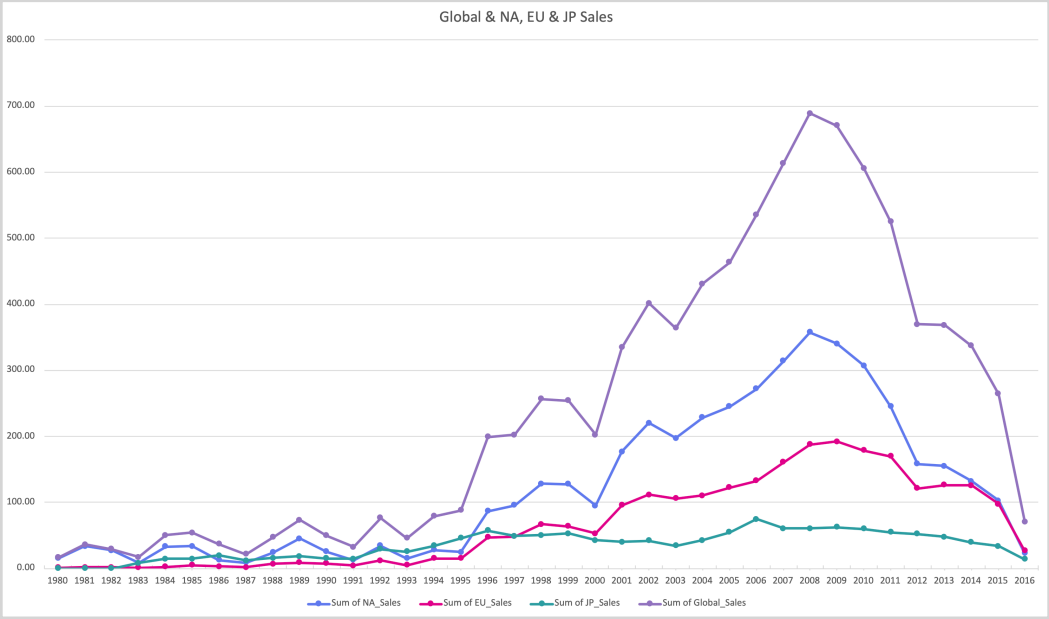
Insights and Implications:
The analysis revealed that **regional sales patterns have shifted significantly**. Assuming stable patterns for 2017 could misallocate the marketing budget, **potentially reducing ROI**. Instead, a region-specific strategy is recommended:

- **North America:** Focus on avoiding market saturation and maintaining quality, essential to avoid volatility.
- **Japan:** Leverage stable demand with innovative console and arcade releases.
- **Europe:** Invest strategically in emerging console and home-computer markets.

Visualisations:

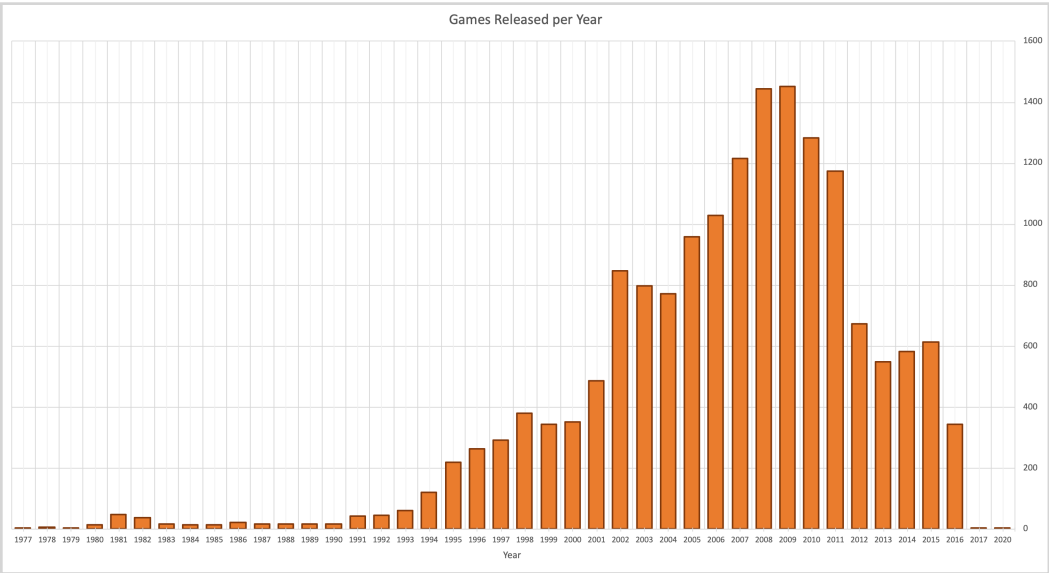
1. Line Chart – Regional Sales Over Time
Purpose: Shows changes in regional sales dominance from 1977 to 2020, highlighting shifts in market leadership.
Description: North America dominates early years, Japan grows strongly in the late 1980s–1990s,

and Europe steadily increases. Recent years show roughly equal contributions from all three regions.



2. PivotTable – Number of Games Released Per Year

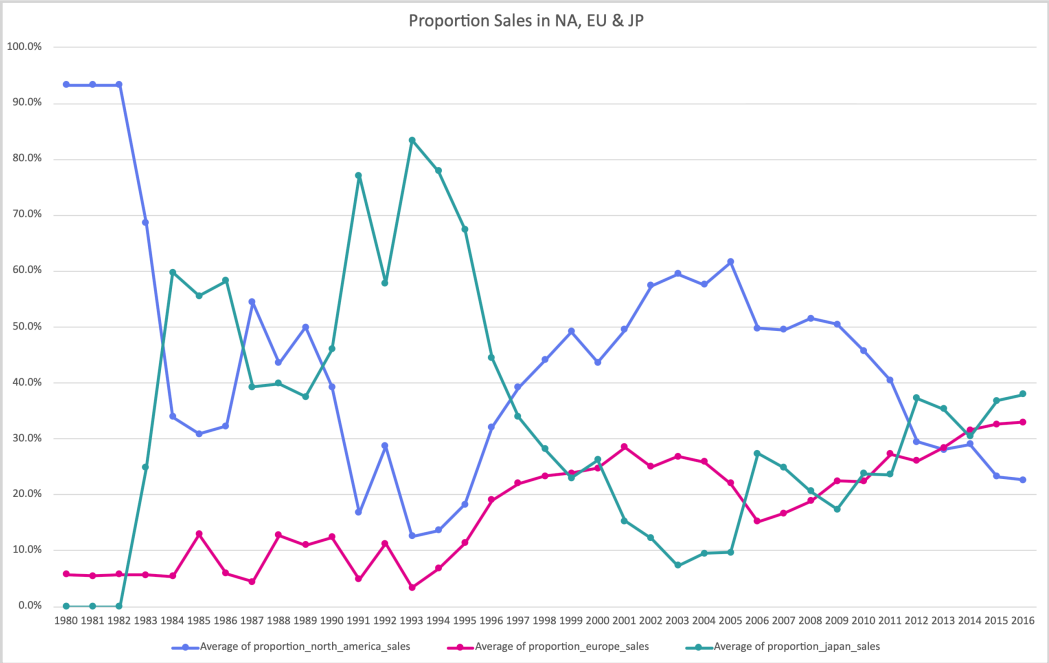
Purpose: Displays production volume trends, highlighting sparse years excluded from analysis.
Description: Early years (1977–1980) and recent years (2017, 2020) show very few releases. The peak period (~1995–2016) shows hundreds of game releases per year, reflecting market growth and activity levels.



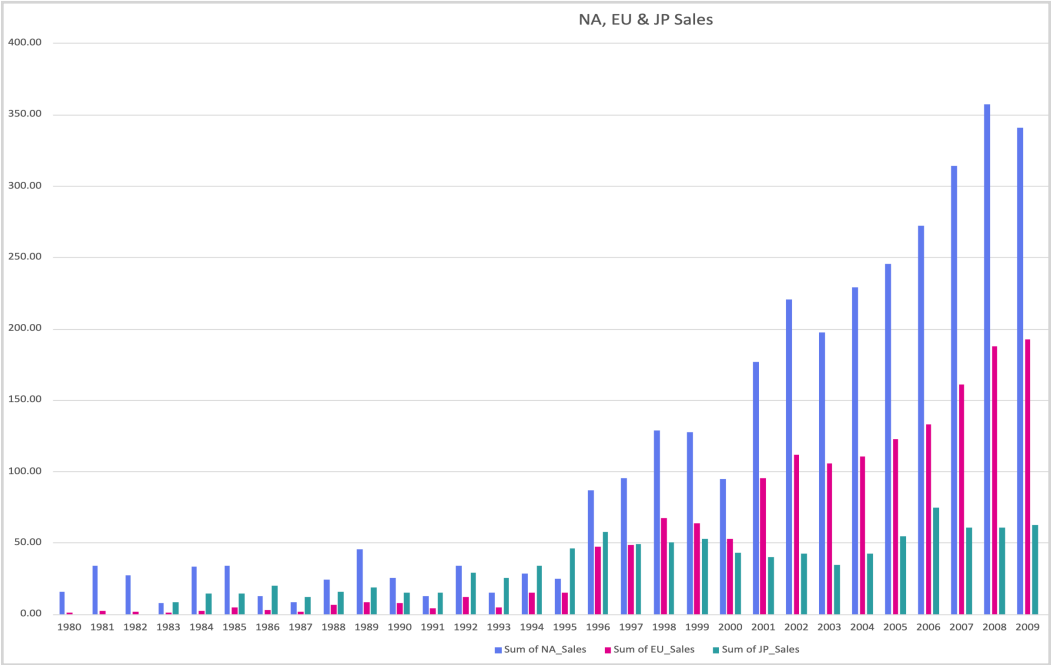
Refining visualisations for the final presentation

For the final presentation to GameCo executives, I refined the **analytical visualisations** used during exploration into simpler, high-level charts focused on strategic insights. While PivotTables and detailed trend charts were essential during analysis to assess data completeness, exclude sparse years, and identify long-term patterns, they were not suitable for an executive audience due to their complexity.

The primary visualisation selected is a **line chart showing regional sales as a percentage of global sales over time** for North America, Europe, and Japan. This chart is the most suitable choice because it directly addresses the executives' assumption that regional sales patterns have remained stable. Presenting sales as percentage share rather than absolute values makes regional comparisons intuitive and highlights shifts in market dominance. The chart clearly shows North America's early leadership, Japan's strong growth during the late 1980s and 1990s, and the recent convergence toward similar regional shares. This insight is critical for informing the 2017 marketing budget allocation.



To support this, a **bar chart comparing regional sales totals for selected representative years** was included. By excluding incomplete or low-volume years, this visual provides a clear and focused comparison of regional performance at key points in time. Bar charts are effective for executives because they enable quick, side-by-side comparisons without requiring detailed interpretation.



These final visualisations differ from the earlier analytical visuals by being streamlined, filtered, and narrative-driven. They build directly on the exploratory analysis but remove unnecessary detail, ensuring clarity and relevance. Together, they effectively communicate that regional sales patterns have changed over time and that GameCo should adopt a region-specific marketing strategy to maximise return on investment.